

Towards Personalized User Interface Design For News Chatbots: A Pilot Study

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ABSTRACT

Numerous mainstream news organizations have adopted chatbots as an emerging channel for delivering personalized news services. However, the news presentation style of chatbots may not satisfy users' personalization needs due to the "one-size-fits-all" design. In this context, we conducted two within-subject studies to investigate the impacts of news reader types, namely "dipper", "tracker", and "reviewer" on users' preference for different interfaces for news selection and reading. Our preliminary findings reveal that all these types of news readers perceived a higher level of ease of selection when using the carousel design. For "trackers" and "reviewers", the question-driven design resulted in a higher level of perceived ease of reading than the in-conversation design. Finally, I discuss my findings and provide insights into personalized user interface design for news chatbots.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**; **User studies**.

KEYWORDS

news chatbot, conversational agent, personalized interface, empirical study

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1 INTRODUCTION

In this digital media era, news organizations are committed to providing personalized news feeds to engage their audiences. Recently, conversational agents or chatbots have become an emerging medium that several mainstream media organizations, such as ABC News, BBC News, and NBC News, have employed to deliver and recommend news stories to their respective audiences[15]. As stated

by the BBC News Lab [7], "news chatbots exemplify a rise in 'conversational journalism' approaches that aim to engage people who are traditionally not users of news through the leveraging of new technologies to create formats that are informal, interactive, and novel." Given the conversational communication style, news chatbots offer news organizations the ability to deliver tailored services and interactive news content [9, 13].

Online news reading is a highly individual activity, and individuals may engage with news unconsciously or purposefully. Previous research has shown that online news consumption behavior varies dramatically among users, primarily in the two stages of consuming news: *selection* and *reading* [5]. Given diverse news consumption behaviors [1, 2], researchers have proposed various methods to provide personalized news content based on personal and behavioral features such as user interests [11], clicking behavior [12], and contexts [6]. However, personalized news content may not always satisfy users' personalization needs due to the "one-size-fits-all" approach in news presentation. In such cases, Constantinides et al. [4] have collected users' features (e.g., reading style and reading time) of news interactions and clustered three stereotypes of news readers: "dippers", "trackers", and "reviewers". Based on these findings, they have proposed a framework [3] to model users' reading habits and give design suggestions to provide personalized interfaces. Nevertheless, these design strategies of graphical user interfaces (GUIs) for mobile applications may not be entirely applicable to conversational user interfaces (CUIs) on mobile devices because CUIs can deliver news content through multi-turn conversations, which requires a new design consideration for interacting with news content.

To explore the design space for personalized user interfaces of news chatbots, we conducted two within-subject studies to compare three practical user interfaces (UIs) for news selection (Study 1) and news reading (Study 2) and to investigate whether and how these UIs should be adapted to different types of news readers. Specifically, Study 1 compared three design alternatives for news selection (i.e., single card, carousel, and list), and Study 2 investigated three design alternatives for news reading (i.e., pop-up webview, in-conversation, and question-driven). We selected most of our design alternatives based on the review of existing news chatbots (e.g., the ABC News chatbot ¹ the NBC News chatbot ², and the La Jornada chatbot ³).

These pilot studies answered two research questions (RQs):

RQ1: Whether news reader types influence the preference for interface design when selecting news in a news chatbot?

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¹m.me/abcnews.au

²m.me/nbcnews

³m.me/lajornadaonline

RQ2: Whether news reader types influence the preference for interface design when reading news in a news chatbot?

The analysis results reveal that users' preferences for UIs depend on the types of news readers. During the selection stage, the carousel design generally led to higher ease of selecting news compared to the single card and list design. However, "trackers" preferred the carousel design while "dippers" preferred the list design. At reading the stage, the majority of users preferred the pop-up webview design while "trackers" also showed a similar preference for the question-driven design. Conversely, the in-conversation design resulted in lower ease of reading news than the question-driven design among "trackers" and "reviewers".

To the best of our knowledge, no existing work has studied personalized interfaces of news chatbots in the news selection and reading process. Therefore, our work provides a comparison of interface design alternatives for news chatbots and investigates the impact of news reader types on user interaction with a news chatbot. We also provide several implications and believe that our empirical studies can provide practical insights for the design of personalized news chatbot interfaces.

2 STUDY 1: DESIGN ALTERNATIVES FOR NEWS SELECTION

To investigate the preferences of different types of news readers for three news selection interface design alternatives, we conducted a pilot study. In this section, we introduce the interface designs, describe the experimental procedure, and report the results of our analysis.

2.1 Interface Design

In the news selection stage, users interact with the chatbot to select the news stories they want to read. We identified three typical design alternatives for this stage. The first is the *single card* design, which presents a news card (Fig. 1 (a)) in a single-turn interaction, similar to the ABC news chatbot⁴. The news card includes the news title, a brief abstract, and a relatively large news thumbnail. Readers can tap the "Read the story" button to read the news or the "Next story" button to skip to the next news. In the *card carousel* design (Fig. 1 (b)), three news cards follow the same format and are displayed at once, allowing readers to swipe left and right to browse them. The *list* design presents (Fig. 1 (c)) three news titles with relevant small topic thumbnails in a list format, and readers can tap the title to read the news in detail.

2.2 Participates

We recruited participants from the popular crowd-sourced platform Prolific⁵ under the approval of the Research Ethics Committee of the author's university. To control for the potential effects of language and cultural background, we established recruitment criteria that participants had to be of UK nationality, currently residing in the UK, and have English as their first language, given that the news presented in this study was collected from mainstream UK media. Thirty-nine participants completed this experiment, with an average completion time of 7 minutes and 46 seconds. Each participant

was compensated with £0.98 after they completed the experiment. To ensure high data quality, we filtered out three participants whose completion time was excessively long (i.e., larger than three times the interquartile range). The final sample consisted of 36 participants (23 women and 13 men), 30.56% of whom ($N = 11$) were 18 to 30 years old, 50.00% ($N = 18$) were aged 31 to 50, and 19.44% ($N = 7$) were older than 50. Regarding educational level, 36.11% of the participants ($N = 13$) had a bachelor's degree or higher.

2.3 Procedure

At the beginning of the experiment, participants were required to indicate their demographic information and select a news topic they wanted to browse after they filled in the informed consent. We provided six news topics (business, entertainment, health, science, sports, and technology), each with nine news stories from mainstream UK news organizations (e.g., BBC, The Telegraph, and Daily Mail). We selected these topics since these soft news articles are generally value-neutral, relevant to a wide audience, and easy to understand [14]. Participants then read the instructions to ensure they understood the operation of each interface. We followed the counterbalanced within-subject design that randomized the order of the three news selection interfaces. To capture participants' natural browsing behavior, we did not limit their browsing time or the number of news stories they could read. Each interface was followed by a question, "This browsing style allows me to select a news article easily [4]". Participants were asked to provide the perceived ease of selecting news using that interface on a 7-point scale. After they explored all three interfaces, we asked them to choose their preferred interface and identify which type of news reader they were (i.e., tracker, reviewer, and dipper) based on the detailed description of each reader type provided. For example, a "tracker" was described as someone who "has a casual interest in the news but likes to read news on specific topics such as sport" [4].

2.4 Results

In general, the number of user preferences for each of the three interfaces was similar, with 36.11% preferring the carousel design ($N = 13$), 33.33% preferring the list design ($N = 12$), and 30.56% preferring the list design ($N = 11$). However, preferences varied greatly between different types of readers. As shown in Fig. 2, over half of the "trackers" (60.00%) preferred the carousel design, while nearly half of "dippers" (46.67%) chose the list design. However, "reviewers" did not show an obvious tendency to favor a particular user interface.

To test the differences among the three types of readers regarding their preferences for the interface preference, we performed a chi-square goodness-of-fit test, which did not show statistical significance. Furthermore, we conducted a one-way repeated measures ANOVA to compare the three user interfaces regarding the perceived ease of news selection. For each news reader group, we found a significant effect of the design alternatives on perceived ease of selection (Dipper: $F(2, 28) = 3.81, p < .05$; Tracker: $F(2, 18) = 4.50, p < .05$; Reviewer: $F(2, 20) = 8.69, p < .01$). The post-hoc paired t-test among these three news reader groups indicated the same pattern: users perceived significantly higher ease of selection under the carousel condition than the single card or

⁴m.me/abcnews.au

⁵<https://www.prolific.co/>

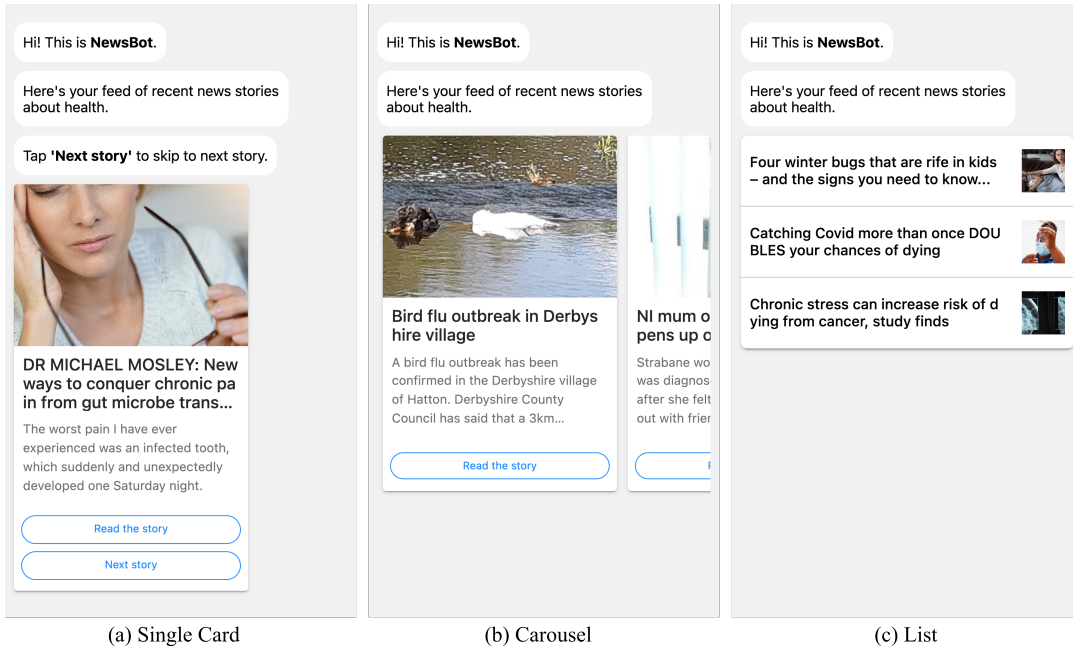


Figure 1: Screenshots of the three interfaces for news selection.

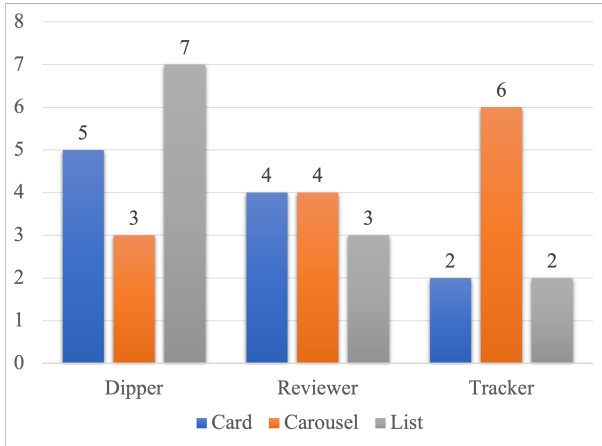


Figure 2: The number of preferences per news selection interface categorized by news reader type.

list condition. However, our analysis did not discover any significant difference regarding news selection between the card and list conditions. Table 1 shows the mean and standard deviation (SD) of perceived ease of selection. The results suggest that the carousel design was the best design for news selection.

3 STUDY 2: DESIGN ALTERNATIVES FOR NEWS READING

In Study 2, we continue to use a within-subjects design experiment to compare three user interfaces for news reading and to investigate the effects of reader type on their choices of user interfaces. The

Table 1: Descriptive Analysis of Perceived Ease of Selection

	Single Card Mean (SD)	Carousal Mean (SD)	List Mean (SD)
Dipper	4.47 (1.30)	5.60 (1.35)	4.67 (1.84)
Reviewer	4.27 (1.95)	6.27 (1.01)	4.09 (1.58)
Tracker	3.80 (2.44)	6.20 (0.92)	4.90 (2.18)

experimental procedure and participant recruitment were similar to the pilot Study 1.

3.1 Interface Design

Fig. 3 demonstrates the three design candidates used in this study for news reading. The “*pop-up webview (PW)*” design is a common format where each news item is displayed in a pop-up window (Fig. 3 (a)). This design can maintain the original style of the news webpage for easy integration into the chatbot, but the pop-up window may interrupt the conversation. For the “*in-conversation (IC)*” design, the news title, author, thumbnails, and each paragraph of news content are directly displayed in chat bubbles (Fig. 3 (b)). Users can tap the “Next” button to read each paragraph sequentially. This simple design integrates news content directly into the conversation, making news delivery more conversational. The “*question-driven (QD)*” design, loosely inspired by the work of Philippe, John, and Marti [10] on a question-driven news chatbot, presents the news content in a Question and Answer (Q&A) format. The chatbot presents a question list first, and each question corresponds to several paragraphs of news content (Fig. 3 (c)). Participants can view the content by tapping these question buttons.

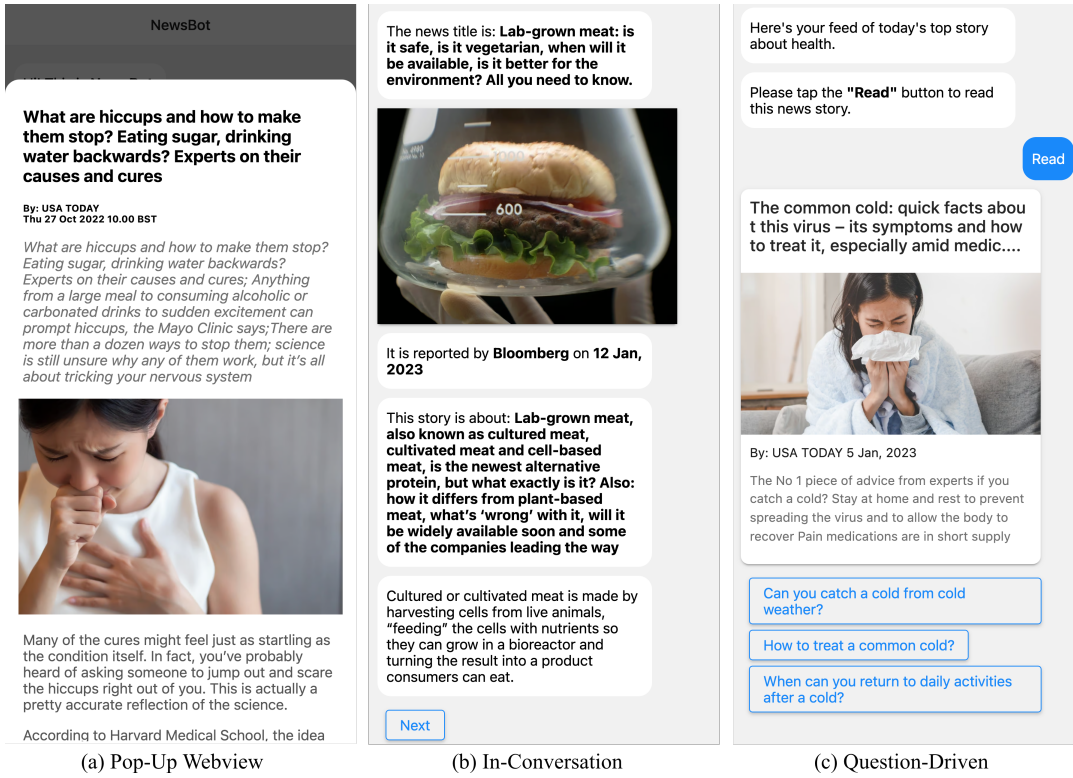


Figure 3: Screenshots of these three interfaces for news reading.

3.2 Procedure

The procedure for this study followed a similar routine to Study 1. We utilized counter-balancing by alternating the presentation interface order for news reading to eliminate the order effect. Additionally, to avoid the influence of news content, we selected nine health-related news items of similar length. Participants were allowed to select one of three news stories they wanted to read before entering an interaction with the candidate interface. After experiencing an interface, participants were asked to indicate their perceived ease of reading in terms of the interface by answering the question, “The layout of this news chatbot makes reading news easy,” using a 7-point scale. This question was adapted from a previous study [4]. Finally, we asked participants to indicate which interface they preferred for reading news and what type of news reader they identified as.

3.3 Participants

A total of 40 users participated in this study. We pre-screened participants with the same criteria as pilot Study 1. The outlier filtering of experimental participation duration did not filter out any participants. In other words, we had 40 valid samples, with 31 females and 9 males. 30.00% of them ($N = 12$) are 18 to 30 years old, 55.00% of them ($N = 22$) are aged 31 to 50, and 15.00% ($N = 6$) are older than 50. Over half of the participants ($N = 22$) have a bachelor’s degree or above. On average, it took about 10

minutes and 46 seconds to complete the experiment. Participants were compensated with £1.23 for their time and participation.

3.4 Results

Fig. 4 illustrates the number of participants who preferred the certain news reading interface, grouped by their type of news reader. Overall, the pop-up webview was the most popular design, with 45.00% ($N = 18$) of participants selecting it, followed by the question-driven design at 32.50% ($N = 13$). Lastly, the in-conversation design at 22.50% ($N = 9$). Specifically, over half of the “reviewers” (54.55%) preferred the pop-up webview design. For the “trackers” group, the pop-up webview and the question-driven designs were equally preferred, while only 15.78% of this group preferred the in-conversation design. However, no clear preference for a particular interface was observed among the “dipper” group.

A chi-square goodness-of-fit test was performed to further investigate users’ interface preferences, but no significant effects were found. A one-way repeated measures ANOVA was conducted, which did not reveal a significant effect of interface design on perceived ease of reading. However, post-hoc paired t-test showed marginally significant differences, with both the “reviewers” group ($F(2, 16) = 2.36, p = 0.09$) and “trackers” group ($F(2, 24) = 2.11, p = 0.06$) reporting higher levels of perceived ease of reading with the question-driven design compared to the in-conversation design. Table 2 presents the mean and standard deviation (SD) of perceived ease of reading for each interface, indicating that the in-conversation design was the least recommended.

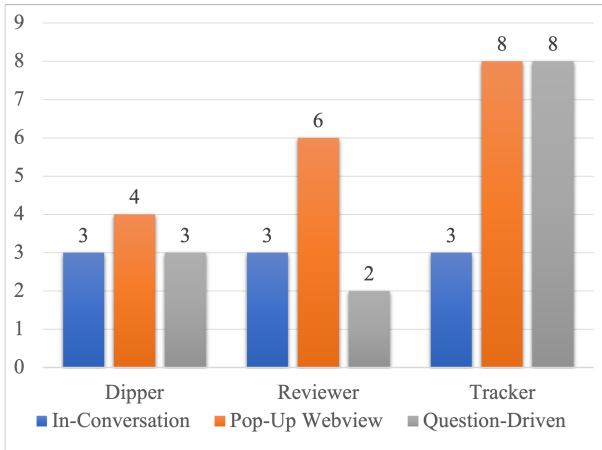


Figure 4: The number of preferences per news reading interface categorized by news reader type.

Table 2: Descriptive Analysis of Perceived Ease of Reading

	IC Mean (SD)	PW Mean (SD)	QD Mean (SD)
Dipper	4.40 (2.59)	5.00 (2.11)	5.30 (1.34)
Reviewer	2.91 (1.87)	5.18 (1.54)	4.55 (1.69)
Tracker	4.37 (1.74)	5.42 (1.39)	5.47 (1.95)

4 DISCUSSION

Based on these two pilot studies, we compared different user interface designs for selection and reading news and investigated personalized user interface design based on reader types for a news chatbot. In this section, we will discuss our results in relation to our research questions and provide practical suggestions for news chatbot designers.

RQ1: Whether news reader types influence the preference for interface design when selecting news in a news chatbot?

The first study’s results suggest that the carousel design was the most preferred interface for news selection across the three news reader types. A possible reason is that this design presents more information than the single card and the list design in a single round of interaction, which allows users to browse the news and make decisions with less effort during the interaction. While there was no significant difference in preference for the user interface among reader types, the differences in preference distributions suggest that news reader types may influence interface preferences. For instance, the carousel design, which shows multiple news items with abstracts, may cater to the efficiency needs of “trackers” who want to quickly get updates on the news they are following, while the list design displays several news items with highlighted titles with few interaction requirements that may better suit “dippers” to find news stories on a specific topic they like. “Reviewers” showed a close preference for all three interfaces, possibly because they like to dive deep into news stories and spend more time reading

them, the interface design differences for news selection may be less likely to have a perceivable impact on their news consumption.

RQ2: Whether news reader types influence the preference for interface design when reading news stories in a news chatbot?

These two significant differences between the in-conversation and the question-driven designs suggest that integrating news content into conversations with news chatbots requires additional design considerations. This is because the simple linear presentation (in-conversation design) had a negative effect on the perceived ease of reading compared to the structured presentation (question-driven design). This result echoes previous research that shows structured and interactive news presentation design has a positive effect on using a new chatbot [9].

Moreover, the pop-up webview design seemed to be a middle-range choice among different types of readers. The user preference distribution shows that “reviewers” preferred the pop-up webview design while “trackers” equally liked the pop-up and question-driven design. A possible explanation for these results may be that frequent news readers may be more familiar with the web-based layout to read news stories [8]. Additionally, the question-driven design offered “trackers” the opportunity to read only the parts they like, which may be why they prefer this design. Since “dippers” rarely read news articles, they may not share a similar preference for reading interfaces in news chatbots.

Our results imply that *efficiency* is a key factor users care about when browsing news in a chatbot. We recommend providing more news items in a single round of conversation to reduce users’ effort to explore news items in a chatbot. It would be helpful to provide a personalized news selection interface with various highlights according to news readers’ type, especially for time-sensitive or irregular news readers. At the news reading stage, pop-up webview is a relatively “safe” design. Designers should further consider the structure and interaction style of the news content when presenting it in a chatbot. As the second study results suggest, displaying news content directly in conversations hurts the user experience of reading news. More solutions for embedding news into conversations are worth exploring for providing personalized news services.

The two pilot studies have a small sample size; therefore, we will recruit more participants to validate our preliminary findings in the future. In addition, we have not analyzed user feedback to reveal the deeper reasons for the different interface preferences. We plan to conduct qualitative studies, such as interviewing users, to gain insight into user perspectives and reveal the influence of more personal characteristics on news chatbot interface design. Additionally, to provide personalized news chatbot services, exploring more design alternatives to integrate news into conversations may also be an interesting direction for future research.

5 CONCLUSION

In this study, we investigated the impact of news reader types on the perceptions of interface design alternatives for a news chatbot. We focused on the two stages of news consumption and compared three design alternatives for the news selection stage and three alternatives for news reading stage. The results show that the carousel design resulted in a higher level of ease in selecting news items for

all types of news readers. At the reading stage, the question-driven design is superior to the in-conversation design for “trackers” and “reviewers” regarding ease of reading. Based on these results, we proposed design implications for creating a personalized interface for news chatbots. We believe that our contribution can complement existing efforts to personalized interfaces and provide insights into the personalized interaction experience of news chatbots.

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REFERENCES

- [1] Li Chen, Zhirun Zhang, Xinzhi Zhang, and Lehong Zhao. 2022. A Pilot Study for Understanding Users’ Attitudes Towards a Conversational Agent for News Recommendation. In *Proceedings of the 4th Conference on Conversational User Interfaces*. 1–6.
- [2] Marios Constantinides and John Dowell. 2016. User interface personalization in news apps. In *CEUR Workshop Proceedings*, Vol. 1618. CEUR workshop Proceedings.
- [3] Marios Constantinides and John Dowell. 2018. A framework for interaction-driven user modeling of mobile news reading behaviour. In *Proceedings of the 26th Conference on User Modeling, Adaptation and Personalization*. 33–41.
- [4] Marios Constantinides, John Dowell, David Johnson, and Sylvain Malacria. 2015. Exploring mobile news reading interactions for news app personalisation. In *Proceedings of the 17th International Conference on Human-Computer Interaction with Mobile Devices and Services*. 457–462.
- [5] Cagdas Esiyok, Benjamin Kille, Brijnesh-Johannes Jain, Frank Hopfgartner, and Sahin Albayrak. 2014. Users’ reading habits in online news portals. In *Proceedings of the 5th Information Interaction in Context Symposium*. 263–266.
- [6] Florent Garcin, Christos Dimitrakakis, and Boi Faltings. 2013. Personalized news recommendation with context trees. In *Proceedings of the 7th ACM Conference on Recommender Systems*. 105–112.
- [7] Bronwyn Jones and Rhianne Jones. 2019. Public service chatbots: Automating conversation with BBC News. *Digital journalism* 7, 8 (2019), 1032–1053.
- [8] Bahar Kaya, Elaheh Behraves, A Mohammed Abubakar, Omer Sami Kaya, and Carlos Orús. 2019. The moderating role of website familiarity in the relationships between e-service quality, e-satisfaction and e-loyalty. *Journal of Internet Commerce* 18, 4 (2019), 369–394.
- [9] Lukas Köb, Stephan Schlögl, and Ellen Richter. 2022. Chatbots for News Delivery—Investigations into Intrinsic Motivation and User Engagement. In *International Conference on Knowledge Management in Organizations*. Springer, 294–305.
- [10] Philippe Laban, John Canny, and Marti A. Hearst. 2020. What’s The Latest? A Question-driven News Chatbot. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*. Association for Computational Linguistics, Online, 380–387. <https://doi.org/10.18653/v1/2020.acl-demos.43>
- [11] Lei Li, Li Zheng, Fan Yang, and Tao Li. 2014. Modeling and broadening temporal user interest in personalized news recommendation. *Expert Systems with Applications* 41, 7 (2014), 3168–3177.
- [12] Jiahui Liu, Peter Dolan, and Elin Rønby Pedersen. 2010. Personalized news recommendation based on click behavior. In *Proceedings of the 15th international conference on Intelligent user interfaces*. 31–40.
- [13] Theodora A Maniou and Andreas Veglis. 2020. Employing a chatbot for news dissemination during crisis: Design, implementation and evaluation. *Future Internet* 12, 7 (2020), 109.
- [14] Carsten Reinemann, James Stanyer, Sebastian Scherr, and Guido Legnante. 2012. Hard and soft news: A review of concepts, operationalizations and key findings. *Journalism* 13, 2 (2012), 221–239.
- [15] Zhirun Zhang, Xinzhi Zhang, and Li Chen. 2021. Informing the Design of a News Chatbot. In *Proceedings of the 21st ACM International Conference on Intelligent Virtual Agents*. 224–231.